MMDU MULLANA OFFICIAL SYLLABUS 2025-26

Mathematics-I

Complete 4-Unit Lecture Notes & University Solved Examination Bank

Course Code: BMAT-001

Units Covered: Units 1, 2, 3, and 4 (Exhaustive Theory + Solved Problems)

Mathematical Rigor: Publication-grade KaTeX typesetting

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UNIT 1 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 1: Unit I: Matrices (10 Hours)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 35 mins

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1. Rank of a Matrix & Normal (Canonical) Form

1. In-Depth Mathematical Foundations of Matrix Rank

✅ **FUNDAMENTAL LAW / RULE:**
Rigorous Linear Algebra Definition: Let A be an $m \times n$ matrix with entries in $\mathbb{R}$ (or $\mathbb{C}$). The **Rank** of A, denoted by $\rho(A)$ or $\text{rank}(A)$, is defined as the dimension of the vector space spanned by its row vectors (Row Rank) or column vectors (Column Rank). By the Fundamental Rank Theorem of Linear Algebra:

$$\text{Row Rank}(\mathbf{A}) = \text{Column Rank}(\mathbf{A}) = \rho(\mathbf{A}) \leq \min(\mathbf{m}, \mathbf{n})$$

Minor-Based Criterion of Rank:

A matrix A is said to be of rank r if and only if:

1. There exists **at least one minor** (determinant of a square submatrix) of order r that is strictly non-zero: $|M_r| \neq 0$.
2. **Every minor** of order $(r+1)$ and higher vanishes identically: $|M_{r+1}| = 0$.

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2. Deep-Dive: Row Echelon Form Reduction

Strict Mathematical Criteria for Row Echelon Form:

A matrix E is in **Row Echelon Form** if:

1. **Zero Rows at Bottom:** All rows consisting entirely of zeros are grouped at the very bottom.
2. **Leading Entry (Pivot):** The leading coefficient (the first non-zero entry from the left) of each non-zero row is strictly to the right of the leading coefficient of the row directly above it.
3. **Zeros Below Pivots:** All entries in a column directly below a pivot entry are zero.

$$\rho(\mathbf{A}) = \text{Total Number of Non-Zero Rows in Row Echelon Form}$$

Invariance of Rank under Elementary Row Transformations:

Elementary row transformations do not alter the linear span of the row vectors:

1. Interchanging two rows: $R_i \leftrightarrow R_j$ (Permutation matrix P, $|P| = -1 \neq 0$).

2. Multiplying a row by a non-zero scalar: $R_i \to kR_i,\ k \neq 0$ (Diagonal scaling matrix D, $|D| = k \neq 0$).

3. Adding a scalar multiple of one row to another: $R_i \to R_i + cR_j$ (Shear matrix S, $|S| = 1 \neq 0$).

Because all elementary matrices are non-singular ($|E| \neq 0$), rank is strictly invariant: $\rho(EA) = \rho(A)$.

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3. Deep-Dive: Normal (Canonical) Form $[I_r\ 0; 0\ 0]$

By using **both elementary row AND elementary column operations**, any arbitrary $m \times n$ matrix A of rank r can be brought into its unique Canonical Normal Form:

$$\mathbf{N} = \begin{bmatrix} I_r & 0_{r\times(n-r)} \\ 0_{(m-r)\times r} & 0_{(m-r)\times(n-r)} \end{bmatrix}$$

Existence of Non-Singular Matrices P and Q:

There exist non-singular square matrices P (of order m) and Q (of order n) representing the product of all row operations and column operations respectively, such that:

$$\mathbf{P} \cdot \mathbf{A} \cdot \mathbf{Q} = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$$

- **Procedure to compute P and Q simultaneously:**

1. Write $A_{m\times n} = I_m \cdot A \cdot I_n$.
2. Perform elementary row operations on A and simultaneously on the pre-factor I_m.
3. Perform elementary column operations on A and simultaneously on the post-factor I_n.
4. When the central matrix reduces to $[I_r\ 0; 0\ 0]$, the left matrix becomes P and the right matrix becomes Q!

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4. Fundamental Properties & Sylvester's Rank Inequalities

1. **Sub-multiplicative Property:** $\rho(AB) \leq \min(\rho(A), \rho(B))$.

2. **Sylvester's Law of Nullity:** For an $m \times n$ matrix A and $n \times p$ matrix B:

$$\rho(\mathbf{A}) + \rho(\mathbf{B}) - \mathbf{n} \leq \rho(\mathbf{AB}) \leq \min(\rho(\mathbf{A}), \rho(\mathbf{B}))$$

3. **Rank of Transpose & Conjugate:** $\rho(A^T) = \rho(A)$ and $\rho(A^T A) = \rho(AA^T) = \rho(A)$.

4. **Rank of Sum:** $\rho(A + B) \leq \rho(A) + \rho(B)$.

Problem:

Find the rank of matrix $A = \begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix}$.

Step 1 🚀: Compute the Determinant of A

$$|A| = (2)(6) - (4)(3) = 12 - 12 = 0$$

Since $|A| = 0$, the rank cannot be 2: $\rho(A) < 2$.

Step 2 ⚡: Check Order-1 Minors (Single Entries)

The matrix contains non-zero entries (e.g., $a_{11} = 2 \neq 0$). Therefore, there is at least one non-vanishing minor of order 1.

Step 3 🎯: Row Echelon Verification ($R_2 \to R_2 - 1.5R_1$)

$$\begin{bmatrix} 2 & 4 \\ 3 & 6 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 \\ 0 & 0 \end{bmatrix}$$

There is exactly **1 non-zero row**.

🎯 Rank of A: $\rho(\mathbf{A}) = \mathbf{1}$.

University Question (4 Marks):

Reduce the matrix A to Row Echelon form and determine its rank:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$

Step 1 🚀: Eliminate entries below pivot in Column 1

Apply row operations $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2 ⚡: Eliminate entry below pivot in Column 3

Apply $R_3 \to R_3 - R_2$:

$$A \sim \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Step 3 🎯: Count Non-Zero Rows

The echelon form contains 2 non-zero rows ($[1, 2, 3]$ and $[0, 0, 1]$) and 1 zero row.

🎯 Row Echelon Rank: $\rho(\mathbf{A}) = \mathbf{2}$.

University Question (6 Marks - MMDU Annual Exam):

Reduce matrix A to its Normal Form $\begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$ using elementary row AND column transformations, and find non-singular matrices P and Q such that $PAQ = \begin{bmatrix} I_r & 0 \\ 0 & 0 \end{bmatrix}$:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Step 1 🚀: Setup Identity Matrices $A = I_3 A I_3$

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Step 2 ⚡: Elementary Row Operation $R_2 \to R_2 - R_1$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A I_3$$

Step 3 🎯: Elementary Column Operations $C_2 \to C_2 - C_1,\ C_3 \to C_3 - 2C_1$

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 1\\0 & 1 & 1\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\-1 & 1 & 0\\0 & 0 & 1\end{bmatrix} A \begin{bmatrix}1 & -1 & -2\\0 & 1 & 0\\0 & 0 & 1\end{bmatrix}$$

Step 4 🏁: Row Operation $R_3 \to R_3 - R_2$ and Column Operation $C_3 \to C_3 - C_2$

$$\begin{bmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 0\end{bmatrix} = \begin{bmatrix}1 & 0 & 0\\-1 & 1 & 0\\1 & -1 & 1\end{bmatrix} A \begin{bmatrix}1 & -1 & -1\\0 & 1 & -1\\0 & 0 & 1\end{bmatrix}$$

This is the standard Normal Form $\begin{bmatrix}I_2 & 0\\0 & 0\end{bmatrix}$.

🎯 Canonical Normal Form: $\begin{bmatrix}I_2 & 0\\0 & 0\end{bmatrix}$ | Rank $\rho(\mathbf{A}) = \mathbf{2}$ | $P = \begin{bmatrix}1 & 0 & 0\\-1 & 1 & 0\\1 & -1 & 1\end{bmatrix}$, $Q = \begin{bmatrix}1 & -1 & -1\\0 & 1 & -1\\0 & 0 & 1\end{bmatrix}$.

2. Gauss-Jordan Method for Matrix Inverse [A | I] ~ [I | A^{-1}]

1. The Gauss-Jordan Inversion Algorithm (Theoretical Analysis)

📌 **CRITICAL EXAM DERIVATION:**
University Exam Distinction: Never use the classical Adjugate formula $A^{-1} = \frac{\text{adj}(A)}{|A|}$ when the question explicitly states **"Find inverse using Gauss-Jordan method"**! The classical method requires $O(n!)$ operations via minors, whereas Gauss-Jordan requires only $O(n^3)$ row operations.

Mathematical Principle of Gauss-Jordan Inversion:

Let A be an $n \times n$ invertible matrix ($|A| \neq 0$). If a sequence of elementary row operations $E_1, E_2, \dots, E_k$ transforms A into the identity matrix I:

$$(E_k \dots E_2 E_1)A = I \implies \mathbf{A^{-1} = E_k \dots E_2 E_1}$$

Therefore, if we apply the identical sequence of operations to the augmented matrix $[A \mid I_n]$:

$$(E_k \dots E_2 E_1)[A \mid I_n] = [(E_k \dots E_1)A \mid (E_k \dots E_1)I_n] = [\mathbf{I_n \mid A^{-1}}]$$

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2. Step-by-Step Algorithmic Execution Pipeline

1. **Augmentation:** Construct the partitioned block matrix $[A \mid I_n]$ of dimension $n \times 2n$.

2. **Forward Elimination (Upper Triangularization):**

- For column $j = 1, 2, \ldots, n$:
- **Pivoting:** Find row $i \geq j$ with a non-zero pivot a_{ij} (preferring 1 or -1). Swap $R_j \leftrightarrow R_i$.
- If pivot entry $a_{jj} = 0$, matrix is **singular** (A^{-1} does not exist!).
- Normalize the pivot row: $R_j \to \frac{1}{a_{jj}} R_j$.
- Eliminate all entries below pivot: for each $k > j$, $R_k \to R_k - a_{kj} R_j$.

3. **Backward Elimination (Diagonalization):**

- Starting from the bottom row $j = n, n-1, \ldots, 2$:
- For each row $k < j$, eliminate the above-pivot entries: $R_k \to R_k - a_{kj} R_j$.

4. **Extraction:** The right half of the augmented matrix is strictly A^{-1}. Verify that $A \cdot A^{-1} = I_n$.

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3. Comparison of Matrix Inversion Techniques

Feature	Gauss-Jordan Method	Adjugate Formula	LU Decomposition
Operation Count	n^3 additions & multiplications	$\sim (n+1)!$ determinant operations	$\frac{2}{3}n^3$ operations
Memory Required	$n \times 2n$ augmented grid	High cofactor matrix cache	$n \times n$ in-place factorization
Suitability	Hand calculation in university exams	2×2 matrices only	High-performance computer algorithms

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Problem:

Find the inverse of matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ using the Gauss-Jordan method.

Step 1 🚀: Setup Augmented Matrix $[A \mid I]$

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array}\right]$$

Step 2 ⚡: Make entry below pivot zero ($R_2 \to R_2 - 3R_1$)

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array}\right]$$

Step 3 🎯: Normalize pivot of row 2 ($R_2 \to -\frac{1}{2}R_2$)

$$\sim \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

Step 4 🏁: Eliminate entry above pivot ($R_1 \to R_1 - 2R_2$)

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & \frac{3}{2} & -\frac{1}{2} \end{array}\right]$$

🎯 Final Inverse: $\mathbf{A^{-1}} = \begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix}$. Verified: $A \cdot A^{-1} = I$.

University Question (6 Marks - MMDU Question Paper):
Employ the Gauss-Jordan method to determine the inverse of:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Step 1 🚀: Swap Rows to Get Leading Pivot 1 ($R_1 \leftrightarrow R_2$)

$$[A \mid I] = \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$$

Step 2 ⚡: Eliminate Column 1 ($R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - R_1$)

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & -3 & -1 & 1 & -2 & 0 \\ 0 & -1 & 1 & 0 & -1 & 1 \end{array}\right]$$

Step 3 🎯: Swap $R_2 \leftrightarrow R_3$, Normalize and Eliminate

$$
\begin{aligned}
R_2 \to -R_3: &\quad [0,1,-1 \mid 0,1,-1] \\
R_3 \to R_3 + 3R_2: &\quad [0,0,-4 \mid 1,1,-3] \\
R_3 \to -\frac{1}{4}R_3: &\quad [0,0,1 \mid -\frac{1}{4},-\frac{1}{4},\frac{3}{4}]
\end{aligned}
$$

Step 4 🏁: Eliminate Above Pivots to reach $[I \mid A^{-1}]$

$$
\begin{aligned}
R_2 \to R_2 + R_3: &\quad [0,1,0 \mid -\frac{1}{4},\frac{3}{4},-\frac{1}{4}] \\
R_1 \to R_1 - R_3 - 2R_2: &\quad [1,0,0 \mid \frac{3}{4},-\frac{1}{4},-\frac{1}{4}]
\end{aligned}
$$

🎯 Complete Inverse: $\mathbf{A^{-1}} = \frac{1}{4}\begin{bmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{bmatrix}$. Verified: $|A| = 4 \neq 0$.

3. Consistency of System of Linear Equations AX = B

1. Comprehensive Theory: System of Linear Equations $AX = B$

A system of m linear equations in n unknowns $x_1, x_2, \ldots, x_n$ is represented in matrix notation as:

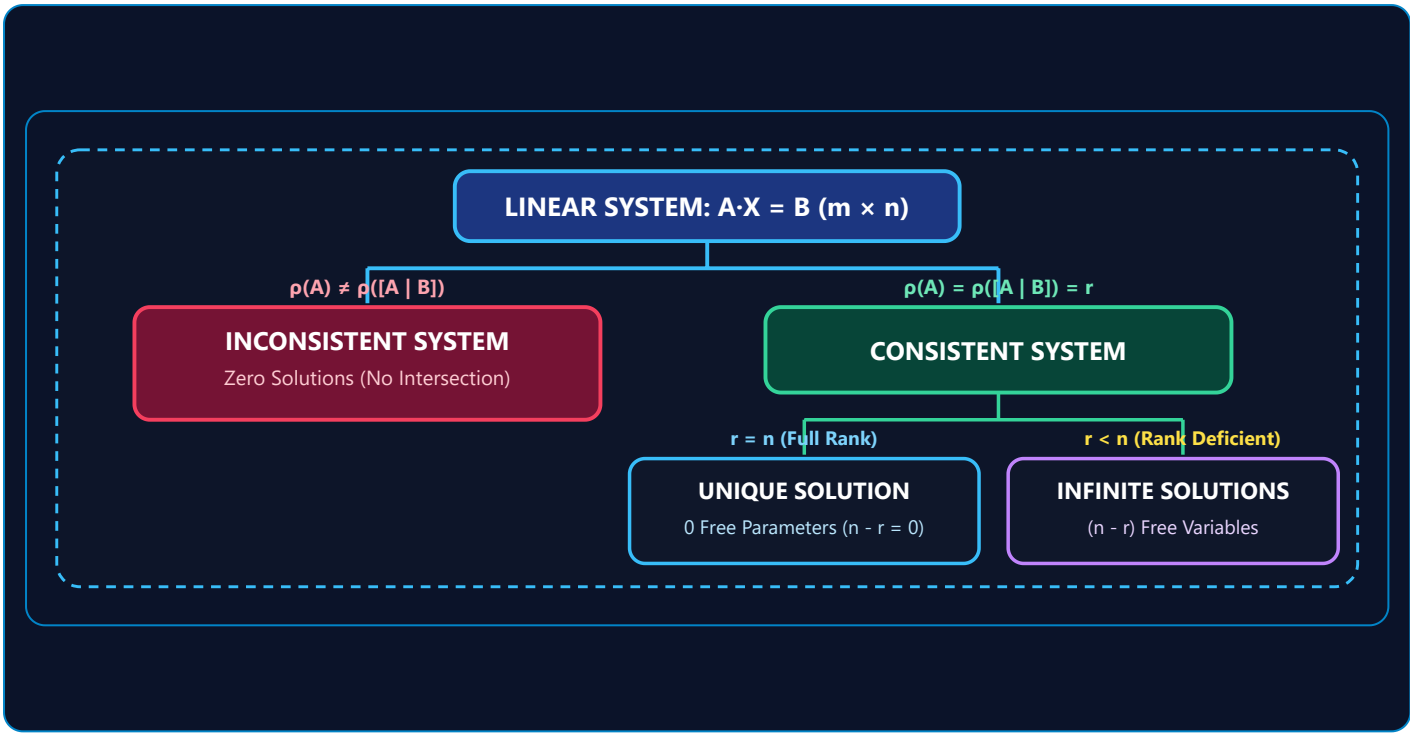
$$
\begin{bmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \iff \mathbf{AX = B}
$$

The Rouché–Capelli Theorem:

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: A system of linear equations $AX = B$ is **consistent** (possesses at least one solution) if and only if the rank of the coefficient matrix A equals the rank of the augmented matrix $[A \mid B]$:

$$
\rho(\mathbf{A}) = \rho([\mathbf{A} \mid \mathbf{B}])
$$

2. Complete Classification of Solution Regimes



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3. Homogeneous Linear Systems $AX = 0$

For $AX = 0$, $B = 0$. Since augmenting a column of zeros cannot increase matrix rank, $\rho(A) = \rho([A \mid 0])$ always holds. Therefore, **a homogeneous system is ALWAYS consistent** and always possesses at least the trivial solution $X = 0$.

Nature of Solutions for Square Systems ($n \times n$):

1. **Trivial Solution ($X = 0$):**

- Occurs when $\rho(A) = n \iff |A| \neq 0$ (Non-singular matrix).
- The columns of A are linearly independent; the only solution is $x_1 = x_2 = \cdots = x_n = 0$.

2. **Non-Trivial Solutions (Infinite Non-Zero Solutions):**

- Occurs when $\rho(A) = r < n \iff |A| = 0$ (Singular matrix), yielding $(n - r)$ linearly independent parameters.

> 🚨 **EXAMINER TRAP / PITFALL:**
> **Examiner's Trap:** When an exam problem asks: "*Find values of k for which the homogeneous equations have non-trivial (non-zero) solutions*", simply set the determinant to zero: $\mathbf{|A| = 0}$!

University Question (6 Marks - Classic MMDU Exam Problem):
Investigate for what values of λ and μ the system of equations:

$$\begin{cases} x+y+z=6 \\ x+2y+3z=10 \\ x+2y+\lambda z=\mu \end{cases}$$

has: (i) No solution, (ii) A unique solution, (iii) Infinitely many solutions.

Step 1 🚀: Setup Augmented Matrix and Echelon Reduction

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 10 \\ 1 & 2 & \lambda & \mu \end{array}\right] \xrightarrow[R_3 \to R_3 - R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 1 & \lambda-1 & \mu-6 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \lambda-3 & \mu-10 \end{array}\right]$$

Step 2 ⚡: Case Analysis by Rouché–Capelli Theorem

1. Case (i) No Solution (Inconsistent): $\lambda = 3,\ \mu \neq 10 \implies \rho(A) = 2 \neq \rho([A \mid B]) = 3$
2. Case (ii) Unique Solution: $\lambda \neq 3,\ \forall \mu \implies \rho(A) = \rho([A \mid B]) = 3 = n$
3. Case (iii) Infinite Solutions: $\lambda = 3,\ \mu = 10 \implies \rho(A) = \rho([A \mid B]) = 2 < n\ (z = k)$

🎯 **Verified Result:** (i) No solution for $\lambda = 3, \mu \neq 10$ | (ii) Unique for $\lambda \neq 3$ | (iii) Infinite for $\lambda = 3, \mu = 10$.

4. Eigenvalues, Eigenvectors & Cayley-Hamilton Theorem

1. Eigenvalues (Characteristic Roots) & Eigenvectors

✅ **FUNDAMENTAL LAW / RULE:**

Definition: Let A be an $n \times n$ square matrix. A scalar λ is called an **Eigenvalue** (or characteristic root / latent root) of A if there exists a **non-zero** column vector $X \neq 0$ such that:

$$\mathbf{AX} = \lambda \mathbf{X} \iff (\mathbf{A} - \lambda \mathbf{I})\mathbf{X} = \mathbf{0}$$

> The non-zero vector X is called the **Eigenvector** (or characteristic vector) corresponding to λ.

Characteristic Equation & Matrix Invariants (3×3 Shortcut):

For $AX = \lambda X$ to have a non-trivial solution $X \neq 0$, the matrix $(A - \lambda I)$ must be singular:

$$\mathbf{|A - \lambda I| = 0}$$

For a 3×3 matrix A, the characteristic polynomial expands directly using trace and principal minors:

$$\mathbf{\lambda^3 - S_1\lambda^2 + S_2\lambda - |A| = 0}$$

Where:

- $S_1 = \text{Tr}(A) = a_{11} + a_{22} + a_{33} = \lambda_1 + \lambda_2 + \lambda_3$ *(Sum of main diagonal elements = Sum of eigenvalues)*
- $S_2 = M_{11} + M_{22} + M_{33}$ *(Sum of cofactors/minors of main diagonal elements)*
- $|A| = \det(A) = \lambda_1 \cdot \lambda_2 \cdot \lambda_3$ *(Determinant = Product of eigenvalues)*

Key Properties of Eigenvalues:

1. The eigenvalues of an **upper or lower triangular (or diagonal) matrix** are simply its main diagonal elements.
2. The eigenvalues of a **real symmetric matrix** ($A^T = A$) are always **purely real**, and eigenvectors of distinct eigenvalues are mutually **orthogonal** ($X_i^T X_j = 0$).
3. If λ is an eigenvalue of A, then:

- λ^k is an eigenvalue of A^k.
- $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} (for $|A| \neq 0$).
- $\frac{|A|}{\lambda}$ is an eigenvalue of $\text{adj}(A)$.

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2. Cayley-Hamilton Theorem & Its Applications

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Theorem Statement: "Every square matrix satisfies its own characteristic equation."**

> If the characteristic equation of an $n \times n$ matrix A is:

$$|A - \lambda I| = (-1)^n (\lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \cdots + c_n) = 0$$

> Then replacing scalar λ by matrix A and constant c_n by $c_n I$ yields the zero matrix:

$$\mathbf{A^n + c_1A^{n-1} + c_2A^{n-2} + \cdots + c_nI = 0}$$

Two Primary Applications of Cayley-Hamilton Theorem:

1. **Finding Matrix Inverse A^{-1} (when $c_n = (-1)^n|A| \neq 0$):**

Multiply the matrix equation by A^{-1} and isolate A^{-1}:

$$\mathbf{A^{-1} = -\frac{1}{c_n}(A^{n-1} + c_1A^{n-2} + \cdots + c_{n-1}I)}$$

2. **Computing Higher Matrix Powers (A^4A^8, etc.):**

Express A^n in terms of lower powers $(A^{n-1}, \ldots, A, I)$ and substitute iteratively.

University Question (6 Marks - MMDU Annual Exam):
For matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$: (1) Verify Cayley-Hamilton Theorem, (2) Evaluate A^{-1}, and (3) Compute A^4.

Step 1 🚀: Characteristic Equation via Invariants $S_1, S_2, |A|$

- $S_1 = \text{Tr}(A) = 1 + 1 + 1 = 3$

- $S_2 = M_{11} + M_{22} + M_{33} = (1-1) + (1-3) + (1-0) = 0 - 2 + 1 = -1$

- $|A| = 1(0) - 2(3) + 1(-3) = -9$

- Characteristic equation: $\lambda^3 - 3\lambda^2 - \lambda + 9 = 0$.

Step 2 ⚡ : Verify $A^3 - 3A^2 - A + 9I = 0$

$$A^2 = \begin{bmatrix} 4 & 3 & 0 \\ -3 & 2 & -2 \\ 6 & 4 & 5 \end{bmatrix}, \quad A^3 = A^2 \cdot A = \begin{bmatrix} 4 & 11 & 1 \\ -9 & -2 & -7 \\ 21 & 11 & 7 \end{bmatrix}$$

Substituting into $A^3 - 3A^2 - A + 9I$:

$$\begin{bmatrix} 4-12-1+9 & 11-9-2+0 & 1-0-1+0 \\ -9+9-0+0 & -2-6-1+9 & -7+6+1+0 \\ 21-18-3+0 & 11-12+1+0 & 7-15-1+9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{(Verified!)}$$

Step 3 🎯 : Deduce A^{-1} and A^4

- Multiply by A^{-1}: $A^2 - 3A - I + 9A^{-1} = 0 \implies$

$$A^{-1} = -\frac{1}{9}(A^2 - 3A - I) = \frac{1}{9}\begin{bmatrix} 0 & 3 & 3 \\ 3 & 2 & -1 \\ 3 & -7 & -1 \end{bmatrix}$$

- Multiply by A: $A^4 = 3A^3 + A^2 - 9A$.

🎯 **Verified Result:** Cayley-Hamilton verified ($A^3 - 3A^2 - A + 9I = 0$) | $A^{-1} = -\frac{1}{9}(A^2 - 3A - I)$.

University Solved Examination Questions (Unit 1)

MMDU Official University Solved Examination Questions (Unit 1)

Course: Mathematics I (`BMAT-001`)

Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 1 (1 Mark): Rank of a Matrix & Row Echelon Form

Question:
Define the rank of a matrix and state the rank of a matrix in Row Echelon Form.

Step-by-Step Solution:
The rank $\rho(A)$ of an $m \times n$ matrix A is the order of its largest non-vanishing minor (or the number of linearly independent rows). In Row Echelon Form, $\rho(A)$ equals the **total number of non-zero rows**.

🎯 **Verified Result:** $\rho(A)$ = Number of non-zero rows in echelon form.

📘 Question 2 (1 Mark): Eigenvalues of Upper Triangular Matrix

Question:
Find the sum and product of the eigenvalues of $A = \begin{bmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$.

Step-by-Step Solution:
Since A is upper triangular, its eigenvalues are its diagonal elements: $\lambda_1 = 3, \lambda_2 = 2, \lambda_3 = 5$.

- **Sum of eigenvalues** = $\text{Tr}(A) = 3 + 2 + 5 = 10$.
- **Product of eigenvalues** = $|A| = 3 \times 2 \times 5 = 30$.

🎯 **Verified Result:** Sum = 10, Product = 30.

📘 Question 13 (2 Marks): Inverse via Cayley-Hamilton Theorem

Question:
State Cayley-Hamilton Theorem. If $A^2 - 4A + 3I = 0$, find A^{-1} in terms of A and I.

Step-by-Step Solution:
Every square matrix satisfies its own characteristic equation $|A - \lambda I| = 0$.
Pre-multiplying $A^2 - 4A + 3I = 0$ by A^{-1}:

$$A - 4I + 3A^{-1} = 0 \implies A^{-1} = \frac{1}{3}(4I - A)$$

🎯 **Verified Result:** $A^{-1} = \frac{1}{3}(4I - A)$.

Unit 2: Unit II: Calculus (10 Hours)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 35 mins

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1. Mean Value Theorems & Power Series Expansions

1. In-Depth Analysis of the Three Fundamental Mean Value Theorems

Differential calculus is built on three interconnected foundation stones: Rolle's Theorem, Lagrange's Mean Value Theorem, and Cauchy's Generalized Mean Value Theorem.

A. Rolle's Theorem (1691)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be a real function satisfying:

1. f is **continuous** on the closed interval $[a, b]$.

2. f is **differentiable** on the open interval (a, b).

3. $f(a) = f(b)$.

> Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = 0}$$

- **Geometric Interpretation:** If a continuous smooth curve starts and ends at the same horizontal height, there must be at least one crest or trough where the tangent line is perfectly horizontal (slope = 0).

- **Physical Interpretation:** If a car returns to its starting point after a road trip, its instantaneous velocity must have been zero at least once when turning around.

B. Lagrange's Mean Value Theorem (First MVT)

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let $f : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one point $c \in (a, b)$ such that:

$$\mathbf{f'(c) = \frac{f(b) - f(a)}{b - a}}$$

- **Geometric Interpretation:** There exists at least one interior point c where the instantaneous tangent line is strictly parallel to the chord (secant line) joining the endpoints $(a, f(a))$ and $(b, f(b))$.
- **Alternative Form:** Setting $b = a + h$ and $c = a + \theta h$ with $0 < \theta < 1$:

$$f(a + h) = f(a) + hf'(a + \theta h)$$

C. Cauchy's Generalized Mean Value Theorem

✅ **FUNDAMENTAL LAW / RULE:**
Theorem Statement: Let two functions $f, g : [a, b] \to \mathrm{R}$ be continuous on $[a, b]$, differentiable on (a, b), and $g'(x) \neq 0$ for all $x \in (a, b)$. Then there exists at least one $c \in (a, b)$ such that:

$$\mathbf{\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}}$$

- **Proof Architecture:** Define auxiliary function $\phi(x) = f(x) - \lambda g(x)$. Choose $\lambda = \frac{f(b)-f(a)}{g(b)-g(a)}$ so that $\phi(a) = \phi(b)$. Apply Rolle's theorem to $\phi(x)$ to obtain $\phi'(c) = f'(c) - \lambda g'(c) = 0$.

---

2. Taylor's Theorem with Remainder & Power Series

Taylor's Theorem with Remainder Terms:

If $f(x)$ and its first $(n - 1)$ derivatives are continuous on $[a, a + h]$ and $f^{(n)}(x)$ exists on $(a, a + h)$, then:

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \cdots + \frac{h^{n-1}}{(n-1)!}f^{(n-1)}(a) + R_n$$

Where the remainder R_n can be expressed as:

- **Lagrange's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{n!}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$

- **Cauchy's Form of Remainder:** $\mathbf{R_n = \frac{h^n}{(n-1)!}(1-\theta)^{n-1}f^{(n)}(a+\theta h)}, \quad (0 < \theta < 1)$

Maclaurin's Series Expansion (Centering at $a = 0$):

When $\lim_{n\to\infty} R_n = 0$, the function can be expressed as an infinite power series:

$$\mathbf{f(x) = f(0) + xf'(0) + \frac{x^2}{2!}f''(0) + \frac{x^3}{3!}f'''(0) + \cdots + \frac{x^n}{n!}f^{(n)}(0) + \ldots}$$

Standard Expansion Catalog for Exams (Expansion by Use of Known Series):

1. $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \ldots \quad (\forall x \in \mathrm{R})$

2. $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots \quad (\forall x \in \mathrm{R})$

3. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots \quad (\forall x \in \mathrm{R})$

4. $\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \quad (-1 < x \leq 1)$

5. $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \ldots \quad (|x| < 1)$

Problem:
Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on the interval $[1, 3]$.

Step 1 🚀: Check Rolle's Three Hypotheses

1. $f(x)$ is a polynomial, hence continuous on $[1, 3]$.

2. $f'(x) = 2x - 4$ exists on $(1, 3)$, hence differentiable on $(1, 3)$.

3. Check boundary values:

$$f(1) = 1^2 - 4(1) + 3 = 0$$

$$f(3) = 3^2 - 4(3) + 3 = 9 - 12 + 3 = 0 \implies f(1) = f(3)$$

Step 2 ⚡: Find Stationary Point $c \in (1, 3)$ **where** $f'(c) = 0$

$$f'(c) = 2c - 4 = 0 \implies \mathbf{c = 2}$$

Since $c = 2 \in (1, 3)$, Rolle's Theorem is **completely verified**!

🎯 Rolle's Theorem verified: $\mathbf{c = 2 \in (1, 3)}$.

University Question (4 Marks - MMDU Question Paper):
Expand $f(x) = e^x \cos x$ up to the term containing x^4 using Maclaurin's series (and verify by multiplication of known series).

Step 1 🚀: Compute Successive Derivatives at $x = 0$

- $f(x) = e^x \cos x \implies f(0) = 1$
- $f'(x) = e^x(\cos x - \sin x) \implies f'(0) = 1(1 - 0) = 1$
- $f''(x) = -2e^x \sin x \implies f''(0) = 0$
- $f'''(x) = -2e^x(\sin x + \cos x) \implies f'''(0) = -2(0 + 1) = -2$
- $f^{(4)}(x) = -4e^x \cos x \implies f^{(4)}(0) = -4(1) = -4$

Step 2 ⚡: Substitute into Maclaurin's Formula (or Known Series)

$$e^x \cos x = (1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots)(1 - \frac{x^2}{2} + \frac{x^4}{24} - \dots) = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + \dots$$

🎯 Maclaurin Expansion: $\mathbf{e^x \cos x = 1 + x - \frac{x^3}{3} - \frac{x^4}{6} + O(x^5)}$.

2. Curvature, Centre & Chord of Curvature (Cartesian, Parametric, Polar)

1. Curvature, Radius & Center of Curvature: Comprehensive Theory

Physical Concept of Curvature (κ):

✅ **FUNDAMENTAL LAW / RULE:**

Definition: Let P and Q be two neighboring points on a smooth curve separated by arc length Δs. If the tangent vectors at P and Q make angles ψ and $\psi + \Delta\psi$ with the positive x-axis, the **Curvature** κ is the rate of turning of the tangent line per unit arc length:

$$\kappa = \lim_{\Delta s \to 0} \left| \frac{\Delta \psi}{\Delta s} \right| = \left| \frac{d\psi}{ds} \right|$$

Radius of Curvature (ρ):

The Radius of Curvature ρ is the reciprocal of the curvature:

$$\rho = \frac{1}{\kappa} = \left| \frac{ds}{d\psi} \right|$$

- A straight line has zero bending $\implies \kappa = 0 \implies \rho = \infty$.
- A circle of radius R bends uniformly $\implies \kappa = 1/R \implies \rho = R$.

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2. Analytical Formulas for Radius of Curvature ρ

Coordinate Representation	Curve Equation	Radius of Curvature Formula ρ
Cartesian $y = f(x)$	$y = f(x)$	$\rho = \frac{(1+y_1^2)^{3/2}}{y_2}$ $(y_1 = \frac{dy}{dx},\ y_2 = \frac{d^2y}{dx^2})$
Cartesian $x = g(y)$	$x = g(y)$	$\rho = \frac{(1+x_1^2)^{3/2}}{x_2}$ $(x_1 = \frac{dx}{dy},\ x_2 = \frac{d^2x}{dy^2})$
Parametric	$x = x(t),\ y = y(t)$	$\rho = \frac{(x'^2+y'^2)^{3/2}}{x'y''-y'x''}$
Polar Form	$r = f(\theta)$	$\rho = \frac{(r^2+r_1^2)^{3/2}}{r^2+2r_1^2-rr_2}$ $(r_1 = \frac{dr}{d\theta},\ r_2 = \frac{d^2r}{d\theta^2})$
Pedal Form (p-r)	$p = f(r)$	$\rho = r\frac{dr}{dp}$

---

3. Center of Curvature $(\bar{x}, \bar{y})$ and Chord of Curvature

The circle that shares the same tangent and curvature as the curve at point $P(x, y)$ is the **Osculating Circle (Circle of Curvature)**. Its center $C(\bar{x}, \bar{y})$ is given by:

$$\bar{x} = x - \frac{y_1(1+y_1^2)}{y_2}, \quad \bar{y} = y + \frac{1+y_1^2}{y_2}$$

- **Equation of Circle of Curvature:** $(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$.
- **Evolute:** The locus of the centers of curvature of all points on a curve is called the **Evolute** of that curve.

Chords of Curvature:

- **Chord parallel to x-axis:** $C_x = 2\rho\sin\psi = \frac{2y_1(1+y_1^2)}{y_2}$
- **Chord parallel to y-axis:** $C_y = 2\rho\cos\psi = \frac{2(1+y_1^2)}{y_2}$
- **Chord through the Pole (Polar Curve $r = f(\theta)$):** $C_p = 2\rho\sin\phi = \frac{2r(r^2+r_1^2)}{r^2+2r_1^2-rr_2}$

University Question (6 Marks):
Find the radius of curvature at any point t on the cycloid $x = a(t+\sin t),\ y = a(1-\cos t)$.

Step 1 🚀: Compute First and Second Derivatives w.r.t t

- $x' = \frac{dx}{dt} = a(1+\cos t) = 2a\cos^2(\frac{t}{2})$

- $y' = \frac{dy}{dt} = a\sin t = 2a\sin(\frac{t}{2})\cos(\frac{t}{2})$

- $x'' = -a\sin t, \quad y'' = a\cos t$

Step 2 ⚡ : Denominator $x'y'' - y'x''$ and Numerator $(x'^2 + y'^2)^{3/2}$

$$x'y'' - y'x'' = a^2(1+\cos t) = 2a^2\cos^2(\frac{t}{2})$$

$$(x'^2 + y'^2)^{3/2} = [4a^2\cos^2(\frac{t}{2})]^{3/2} = 8a^3\cos^3(\frac{t}{2})$$

$$\rho = \frac{8a^3\cos^3(t/2)}{2a^2\cos^2(t/2)} = 4a\cos(\frac{t}{2})$$

🎯 Radius of Curvature: $\rho = \mathbf{4a}\cos(\frac{\mathbf{t}}{\mathbf{2}})$.

University Question (6 Marks - MMDU Question Paper):
Find the coordinates of the centre of curvature $(\bar{X}, \bar{Y})$ and the equation of the evolute of the parabola $y^2 = 4ax$.

Step 1 🚀 : Derivatives for $y^2 = 4ax$

$$2yy_1 = 4a \implies y_1 = \frac{2a}{y}, \quad y_2 = -\frac{4a^2}{y^3}$$

Step 2 ⚡ : Centre of Curvature Formula

$$\bar{X} = x - \frac{y_1(1+y_1^2)}{y_2} = x + \frac{y^2 + 4a^2}{2a} = 3x + 2a$$

$$\bar{Y} = y + \frac{1+y_1^2}{y_2} = -\frac{y^3}{4a^2}$$

Step 3 🎯: Eliminate (x, y) to Obtain the Evolute Equation
Substitute $x = \frac{\bar{X}-2a}{3}$ and $y^2 = (4a^2\bar{Y})^{2/3}$ into $y^2 = 4ax$:

$$27a\bar{Y}^2 = 4(\bar{X} - 2a)^3$$

🎯 Centre of Curvature: $(\bar{X}, \bar{Y}) = (3x + 2a, -\frac{y^3}{4a^2})$ | Evolute: $\mathbf{27aY^2 = 4(X - 2a)^3}$.

3. Indeterminate Forms & L'Hospital's Rule

1. Rigorous Analysis of Indeterminate Forms & L'Hospital's Rule

What is an Indeterminate Form?

An algebraic limit expression $\lim_{x\to a} \frac{f(x)}{g(x)}$ is called **indeterminate** if substituting the limiting point $x = a$ fails to yield a well-defined value because the limiting behaviors of the numerator and denominator conflict.

The 7 Classic Indeterminate Forms:

$$\frac{\mathbf{0}}{\mathbf{0}}, \quad \frac{\infty}{\infty}, \quad \mathbf{0 \times \infty}, \quad \infty - \infty, \quad \mathbf{0^0}, \quad \mathbf{1^\infty}, \quad \infty^{\mathbf{0}}$$

2. L'Hospital's Rule Theorem & Derivation

✅ **FUNDAMENTAL LAW / RULE:**
Theorem (Bernoulli & L'Hospital): Let f and g be differentiable on an open interval containing a (except possibly at a). If $\lim_{x\to a} f(x) = 0$ and $\lim_{x\to a} g(x) = 0$ (or both tend to $\pm\infty$), and $g'(x) \not\equiv 0$ near a, then:

$$\lim_{\mathbf{x\to a}} \frac{\mathbf{f(x)}}{\mathbf{g(x)}} = \lim_{\mathbf{x\to a}} \frac{\mathbf{f'(x)}}{\mathbf{g'(x)}}$$

3. Systematic Reduction Strategies for Non-Fractional Forms

1. **Form $0 \times \infty$:** Invert one factor into the denominator: $f(x) \cdot g(x) = \frac{f(x)}{1/g(x)} \implies \frac{0}{0}$.

2. **Form $\infty - \infty$:** Combine over a common denominator using LCM or rationalization.

3. **Exponential Forms ($0^0, 1^\infty, \infty^0$):** Let $y = [f(x)]^{g(x)}$, take $\ln y = g(x)\ln[f(x)]$, evaluate $L = \lim_{x\to a} \ln y$, and recover $\lim y = e^L$.

University Question (4 Marks):
Evaluate $\lim_{x\to 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x}\right)$.

Step 1 🚀: Convert $\infty - \infty$ into $\frac{0}{0}$ via LCM

$$\lim_{x\to 0}\left(\frac{\sin^2 x - x^2}{x^2 \sin^2 x}\right) = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4} \cdot \lim_{x\to 0}\left(\frac{x}{\sin x}\right)^2 = \lim_{x\to 0} \frac{\sin^2 x - x^2}{x^4}$$

Step 2 ⚡: Use Series Expansion $\sin x = x - \frac{x^3}{6} + O(x^5)$

$$\sin^2 x - x^2 = \left(x - \frac{x^3}{6}\right)^2 - x^2 = -\frac{x^4}{3} + O(x^6)$$

$$\lim_{x\to 0} \frac{-\frac{x^4}{3} + O(x^6)}{x^4} = -\frac{1}{3}$$

🎯 Evaluated Limit: $-\frac{1}{3}$.

University Question (6 Marks - MMDU Question Paper):
Evaluate $\lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2}$.

Step 1 🚀: Identify 1^∞ Form & Take Natural Logarithm
Let $y = \lim_{x\to 0} \left(\frac{\tan x}{x}\right)^{1/x^2} \implies \ln y = \lim_{x\to 0} \frac{\ln\left(\frac{\tan x}{x}\right)}{x^2}$.

Step 2 ⚡: Series Expansion of $\tan x$

$$\frac{\tan x}{x} = 1 + \frac{x^2}{3} + O(x^4) \implies \ln\left(\frac{\tan x}{x}\right) = \frac{x^2}{3} + O(x^4)$$

Step 3 🎯: Evaluate Limit and Exponentiate

$$\ln y = \lim_{x\to 0} \frac{\frac{x^2}{3} + O(x^4)}{x^2} = \frac{1}{3} \implies y = e^{1/3} = \sqrt[3]{e}$$

🎯 Limit: $\mathbf{e^{1/3} = \sqrt[3]{e}}$.

4. Beta and Gamma Functions & Their Properties

1. Rigorous Theory of Beta and Gamma Special Functions

A. The Gamma Function (Euler's Integral of Second Kind):

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For any real number $n > 0$, the **Gamma Function** $\Gamma(n)$ is defined as:

$$\boldsymbol{\Gamma(n) = \int_0^\infty x^{n-1} e^{-x}\, dx \quad (n > 0)}$$

Fundamental Properties of $\Gamma(n)$:

1. **Recurrence Relation:** $\boldsymbol{\Gamma(n+1) = n\,\Gamma(n)}$ and $\boldsymbol{\Gamma(n+1) = n!}$ for positive integers.
2. **Base & Half-Integer Values:** $\Gamma(1) = 1$ and $\boldsymbol{\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}}$.

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2. The Beta Function (Euler's Integral of First Kind)

✅ **FUNDAMENTAL LAW / RULE:**
Definition: For $m > 0, n > 0$, the **Beta Function** $B(m, n)$ is defined as:

$$\boldsymbol{B(m, n) = \int_0^1 x^{m-1}(1-x)^{n-1}\, dx}$$

Key Properties and Master Identities:

1. **Symmetry:** $\mathbf{B(m, n) = B(n, m)}$.

2. **Trigonometric Form:**

$$\mathbf{B(m, n) = 2\int_0^{\pi/2} \sin^{2m-1}\theta \cos^{2n-1}\theta \, d\theta}$$

3. **Infinite Domain Form:**

$$\mathbf{B(m, n) = \int_0^{\infty} \frac{t^{m-1}}{(1+t)^{m+n}} \, dt}$$

4. **Beta-Gamma Bridge Relation:**

$$\mathbf{B(m, n) = \frac{\Gamma(m) \cdot \Gamma(n)}{\Gamma(m+n)}}$$

5. **Walli's / Trigonometric Integral Formula:**

$$\int_0^{\pi/2} \sin^{p}\theta \cos^{q}\theta \, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right)\Gamma\left(\frac{q+1}{2}\right)}{2\,\Gamma\left(\frac{p+q+2}{2}\right)}$$

6. **Legendre's Duplication Formula:**

$$\Gamma(m)\,\Gamma\left(m+\frac{1}{2}\right) = \frac{\sqrt{\pi}}{2^{2m-1}}\,\Gamma(2m)$$

University Question (6 Marks - MMDU Question Paper):
Evaluate $\int_0^{\pi/2} \sin^4\theta \cos^6\theta \, d\theta$ using Beta and Gamma functions, and prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$.

Step 1 🚀: Evaluate Trigonometric Integral ($p = 4, q = 6$)

$$I_1 = \frac{\Gamma(\frac{5}{2})\,\Gamma(\frac{7}{2})}{2\,\Gamma(6)} = \frac{(\frac{3\sqrt{\pi}}{4})(\frac{15\sqrt{\pi}}{8})}{2 \times 120} = \frac{3\pi}{512}$$

Step 2 ⚡: Prove $\int_0^1 \frac{dx}{\sqrt{1-x^4}} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$
Substitute $x^4 = u \implies x = u^{1/4},\ dx = \frac{1}{4}u^{-3/4}du$:

$$I_2 = \frac{1}{4}\int_0^1 u^{1/4-1}(1-u)^{1/2-1}du = \frac{1}{4}B(\frac{1}{4},\frac{1}{2}) = \frac{\Gamma(1/4)\Gamma(1/2)}{4\,\Gamma(3/4)} = \frac{\sqrt{\pi}\,\Gamma(1/4)}{4\,\Gamma(3/4)}$$

🎯 Integral Value: $\frac{3\pi}{512}$ | Formula proved via $B(\frac{1}{4},\frac{1}{2})$.

University Solved Examination Questions (Unit 2)

MMDU Official University Solved Examination Questions (Unit 2)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 4 (1 Mark): Geometric Interpretation of Rolle's Theorem

Question:
State the geometric meaning of the point c in Rolle's Theorem.

Step-by-Step Solution:
If a continuous, differentiable curve $y = f(x)$ has equal values at the endpoints $f(a) = f(b)$, there exists at least one interior point $c \in (a, b)$ where $f'(c) = 0$, meaning the **tangent line at** $(c, f(c))$ **is parallel to the** x**-axis**.

🎯 **Verified Result:** Horizontal tangent parallel to x-axis at $x = c$.

📘 Question 14 (2 Marks): Radius of Curvature of Catenary Curve

Question:

Find the radius of curvature ρ for the catenary $y = c\cosh(x/c)$.

Step-by-Step Solution:

Given $y = c\cosh(x/c)$, we have $y_1 = \sinh(x/c)$ and $y_2 = \frac{1}{c}\cosh(x/c)$.
Since $1 + y_1^2 = 1 + \sinh^2(x/c) = \cosh^2(x/c)$:

$$\rho = \frac{(1+y_1^2)^{3/2}}{y_2} = \frac{\cosh^3(x/c)}{\frac{1}{c}\cosh(x/c)} = c\cosh^2(x/c) = \frac{y^2}{c}$$

🎯 **Verified Result:** $\rho = \frac{y^2}{c}$.

📘 Question 22 (6 Marks): Relation Between Beta and Gamma Functions

Question:

State and prove $B(m,n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$ and hence evaluate $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta$.

Step-by-Step Solution:

Using polar coordinates $x = r\cos\theta, y = r\sin\theta$ in $\Gamma(m)\Gamma(n) = 4\int_0^\infty \int_0^\infty e^{-(x^2+y^2)} x^{2m-1} y^{2n-1}\, dx\, dy$ yields $\Gamma(m)\Gamma(n) = B(m,n)\Gamma(m+n)$.
For $\int_0^{\pi/2} \sin^{1/2}\theta \cos^{-1/2}\theta\, d\theta$: here $p = 1/2, q = -1/2$, so:

$$I = \frac{\Gamma(3/4)\Gamma(1/4)}{2\Gamma(1)} = \frac{1}{2}\cdot\frac{\pi}{\sin(\pi/4)} = \frac{\pi}{\sqrt{2}}$$

🎯 **Verified Result:** $\int_0^{\pi/2} \sqrt{\tan\theta}\, d\theta = \frac{\pi}{\sqrt{2}}$.

UNIT 3 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 3: Unit III: Multivariable Calculus (Differentiation) (10 Hours)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 35 mins

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1. Limits, Continuity & Partial Derivatives

1. Multivariable Functions: Limits, Continuity & Partial Derivatives

A. Limit of a Function of Two Variables:

✅ **FUNDAMENTAL LAW / RULE:**

Rigorous ϵ-δ Definition: We write $\lim_{(x,y)\to(a,b)} f(x,y) = L$ if for every $\epsilon > 0$, there exists a $\delta > 0$ such that:

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta \implies |f(x,y) - L| < \epsilon$$

- **Path Independence Principle:** For a limit to exist in R^2, the function must approach the **exact same value** L **along EVERY conceivable path** approaching (a, b).
- **Two-Path Non-Existence Test:** If $f(x,y)$ yields different values along paths $y = mx$ or $y = mx^2$ (i.e., limit depends on m), the limit **strictly DOES NOT EXIST**!

B. Continuity of $f(x,y)$:

A function $f(x,y)$ is continuous at (a,b) if:

1. $f(a,b)$ is defined.
2. $\lim_{(x,y)\to(a,b)} f(x,y)$ exists.
3. $\lim_{(x,y)\to(a,b)} f(x,y) = f(a,b)$.

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2. Partial Derivatives & Higher-Order Derivatives

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \mathbf{f_x(x,y)} = \lim_{\mathbf{\Delta x \to 0}} \frac{\mathbf{f(x+\Delta x, y) - f(x,y)}}{\mathbf{\Delta x}}, \quad \frac{\partial \mathbf{f}}{\partial \mathbf{y}} = \mathbf{f_y(x,y)} = \lim_{\mathbf{\Delta y \to 0}} \frac{\mathbf{f(x,y+\Delta y) - f(x,y)}}{\mathbf{\Delta y}}$$

✅ **FUNDAMENTAL LAW / RULE:**

Clairaut's Symmetry Theorem: If the mixed partial derivatives f_{xy} and f_{yx} are both **continuous** on an open region containing (a,b), then:

$$\frac{\partial^2 \mathbf{f}}{\partial \mathbf{y} \partial \mathbf{x}} = \frac{\partial^2 \mathbf{f}}{\partial \mathbf{x} \partial \mathbf{y}} \iff \mathbf{f_{yx} = f_{xy}}$$

University Question (4 Marks):
Evaluate whether $\lim_{(x,y)\to(0,0)} \frac{x^2y}{x^4+y^2}$ exists.

Step 1 🚀: Approach along straight line paths $y = mx$

$$\lim_{x\to 0} \frac{x^2(mx)}{x^4 + m^2x^2} = \lim_{x\to 0} \frac{mx}{x^2 + m^2} = 0$$

Step 2 ⚡: Approach along parabolic path $y = mx^2$

$$\lim_{x\to 0} \frac{x^2(mx^2)}{x^4 + m^2x^4} = \frac{m}{1 + m^2}$$

Since the limit depends on m, the limit is **not unique** and **does NOT exist**.

🎯 By Path Test ($y = mx^2$), the limit **does NOT exist**.

University Question (6 Marks - MMDU Question Paper):
If $u = \ln(x^2 + y^2)$, prove that u satisfies 2D Laplace's Equation: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.

Step 1 🚀: First Partial Derivatives

$$\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}, \quad \frac{\partial u}{\partial y} = \frac{2y}{x^2 + y^2}$$

Step 2 ⚡: Second Partial Derivatives via Quotient Rule

$$\frac{\partial^2 u}{\partial x^2} = \frac{2(x^2 + y^2) - 2x(2x)}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}, \quad \frac{\partial^2 u}{\partial y^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$$

Step 3 🎯: Sum the Second Derivatives

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = \frac{2y^2 - 2x^2 + 2x^2 - 2y^2}{(x^2 + y^2)^2} = 0$$

🎯 Proved: $\nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ (u is Harmonic).

2. Euler's Theorem on Homogeneous Functions

1. Homogeneous Functions & Euler's Theorem: Rigorous Analysis

✅ **FUNDAMENTAL LAW / RULE:**
Definition: A multivariable function $f(x,y)$ is **homogeneous of degree** n if for any scalar $t > 0$:

$$\mathbf{f(tx, ty) = t^n f(x, y) \iff f(x, y) = x^n \phi\left(\frac{y}{x}\right)}$$

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2. Euler's Theorem on Homogeneous Functions (First & Second Order)

✅ **FUNDAMENTAL LAW / RULE:**
First-Order Euler's Theorem: If $u = f(x,y)$ is a homogeneous function of x and y of degree n with continuous partial derivatives, then:

$$\mathbf{x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u}$$

✅ **FUNDAMENTAL LAW / RULE:**
Second-Order Euler's Formula:

$$\mathbf{x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)u}$$

🚨 **EXAMINER TRAP / PITFALL:**
Examiner's Composite Function Deduction: If $z = f(u)$ is homogeneous of degree n in (x,y), then:

1. $xu_x + yu_y = n\frac{f(u)}{f'(u)} = g(u)$

> 2. $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1]$

University Question (6 Marks - MMDU Question Paper):

If $u = \sin^{-1}\left(\frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}\right)$, prove that $xu_x + yu_y + zu_z = -3\tan u$.

Step 1 🚀: Convert to Homogeneous Function $v = \sin u$

$$v = \sin u = \frac{x + 2y + 3z}{\sqrt{x^8 + y^8 + z^8}}$$

Degree of numerator is 1; degree of denominator is $\sqrt{t^8} = t^4$. Thus v is homogeneous of degree $n = 1 - 4 = -3$.

Step 2 ⚡: Apply 3-Variable Euler's Theorem to v

$$x\frac{\partial v}{\partial x} + y\frac{\partial v}{\partial y} + z\frac{\partial v}{\partial z} = -3v$$

Using $\frac{\partial v}{\partial x} = \cos u \frac{\partial u}{\partial x}$:

$$\cos u\,(xu_x + yu_y + zu_z) = -3\sin u \implies xu_x + yu_y + zu_z = -3\tan u$$

🎯 Result Proved: $\mathbf{x}\frac{\partial \mathbf{u}}{\partial \mathbf{x}} + \mathbf{y}\frac{\partial \mathbf{u}}{\partial \mathbf{y}} + \mathbf{z}\frac{\partial \mathbf{u}}{\partial \mathbf{z}} = -\mathbf{3}\tan \mathbf{u}$.

University Question (6 Marks - MMDU Previous Paper):

If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$.

Step 1 🚀: Define $z = \tan u$ **(Degree** $n = 3 - 1 = 2$**)**

$$xz_x + yz_y = 2z \implies x\sec^2 u\,u_x + y\sec^2 u\,u_y = 2\tan u \implies xu_x + yu_y = \frac{2\tan u}{\sec^2 u} = \sin 2u = g(u)$$

Step 2 ⚡: Second-Order Euler Identity

$$x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = g(u)\,[g'(u) - 1] = \sin 2u\,(2\cos 2u - 1) = \sin 4u - \sin 2u$$

🎯 Second-Order Euler Identity Proved: $\mathbf{x^2 u_{xx} + 2xy u_{xy} + y^2 u_{yy} = \sin 2u(2\cos 2u - 1)}$.

3. Total Derivative, Maxima-Minima & Lagrange Multipliers

1. Total Differential, Chain Rule & Extrema of Multivariable Functions

A. The Total Differential (du) & Total Derivative ($\frac{du}{dt}$):

- **Total Differential:** $du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy + \frac{\partial u}{\partial z}dz$

- **Total Derivative (Single Parameter t):**

$$\frac{\mathbf{du}}{\mathbf{dt}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\mathbf{dx}}{\mathbf{dt}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\mathbf{dy}}{\mathbf{dt}}$$

- **Composite Chain Rule ($x = x(r,\theta),\ y = y(r,\theta)$):**

$$\frac{\partial \mathbf{u}}{\partial \mathbf{r}} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \mathbf{r}} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \mathbf{r}}, \quad \frac{\partial \mathbf{u}}{\partial \theta} = \frac{\partial \mathbf{u}}{\partial \mathbf{x}}\frac{\partial \mathbf{x}}{\partial \theta} + \frac{\partial \mathbf{u}}{\partial \mathbf{y}}\frac{\partial \mathbf{y}}{\partial \theta}$$

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2. Unconstrained Maxima & Minima of Two Variables ($rt - s^2$ Test)

To locate the local maxima, local minima, and saddle points of $z = f(x,y)$:

1. **Stationary Points:** Solve $f_x = 0$ and $f_y = 0$.

2. **Second-Order Partials at (a,b):** $r = f_{xx},\ s = f_{xy},\ t = f_{yy}$ and Discriminant $\mathbf{\Delta = rt - s^2}$:

Condition on $\Delta = rt - s^2$	Condition on $r = f_{xx}$	Nature of Critical Point (a,b)
$\mathbf{rt - s^2 > 0}$	$\mathbf{r < 0}$	**Local Maximum**
$\mathbf{rt - s^2 > 0}$	$\mathbf{r > 0}$	**Local Minimum**
$\mathbf{rt - s^2 < 0}$	Any sign	**Saddle Point**
$\mathbf{rt - s^2 = 0}$	Indeterminate	**Test Fails** (Further investigation required)

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3. Constrained Optimization: Method of Lagrange Multipliers

To optimize $f(x, y, z)$ subject to constraint $\phi(x, y, z) = 0$:

1. Form Auxiliary Lagrangian: $\mathbf{F(x, y, z, \lambda) = f(x, y, z) + \lambda\, \phi(x, y, z)}$.

2. Solve the stationary system:

$$\mathbf{f_x + \lambda\phi_x = 0}, \quad \mathbf{f_y + \lambda\phi_y = 0}, \quad \mathbf{f_z + \lambda\phi_z = 0}, \quad \mathbf{\phi(x, y, z) = 0}$$

University Question (4 Marks):
Find the local maximum and minimum values of $f(x, y) = x^3 + y^3 - 3xy$.

Step 1 🚀: Find Critical (Stationary) Points
$f_x = 3x^2 - 3y = 0 \implies y = x^2$ and $f_y = 3y^2 - 3x = 0 \implies 3x(x^3 - 1) = 0$.
Critical points: $(0, 0)$ and $(1, 1)$.

Step 2 ⚡: Compute Second Partials r, s, t
$r = f_{xx} = 6x, \quad s = f_{xy} = -3, \quad t = f_{yy} = 6y \implies \Delta = rt - s^2 = 36xy - 9.$

Step 3 🎯: Classify Points

1. **At** $(0, 0)$**:** $\Delta = 0 - 9 = -9 < 0 \implies (0, 0)$ is a **Saddle Point**.

2. **At** $(1, 1)$**:** $\Delta = 36 - 9 = 27 > 0$ and $r = 6 > 0 \implies (1, 1)$ is a **Local Minimum** with $f_{\min}(1, 1) = 1 + 1 - 3 = -1$.

🎯 Saddle point: $(0, 0)$ | Local Minimum at $(1, 1)$ with $\mathbf{f_{min} = -1}$.

University Question (6 Marks - MMDU Question Paper):
Find the maximum volume of a rectangular box open at the top, having a given surface area $S = 108 \text{cm}^2$, using Lagrange's method of multipliers.

Step 1 🚀: Objective and Constraint Functions
Maximize $V = xyz$ subject to $\phi(x, y, z) = xy + 2yz + 2zx - 108 = 0$.
Lagrangian: $F(x, y, z, \lambda) = xyz + \lambda(xy + 2yz + 2zx - 108)$.

Step 2 ⚡: Set First Partials to Zero

$$yz + \lambda(y + 2z) = 0, \quad xz + \lambda(x + 2z) = 0, \quad xy + \lambda(2x + 2y) = 0$$

Equating λ gives $x = y = 2z \implies z = \frac{x}{2}$.

Step 3 🎯: Substitute into Constraint $S = 108$

$$x^2 + 2x(x/2) + 2x(x/2) = 3x^2 = 108 \implies x = 6,\ y = 6,\ z = 3 \implies V_{\max} = 6 \times 6 \times 3 = 108 \text{cm}^3$$

🎯 Dimensions: $6\text{cm} \times 6\text{cm} \times 3\text{cm}$ | Maximum Volume: $\mathbf{V_{max} = 108 cm^3}$.

4. Scalar & Vector Functions, Gradient, Divergence & Curl

1. Vector Differential Calculus: Del, Gradient, Divergence & Curl

✅ **FUNDAMENTAL LAW / RULE:**
The Vector Differential Operator Del (∇):

$$\nabla = \hat{\mathbf{i}}\frac{\partial}{\partial x} + \hat{\mathbf{j}}\frac{\partial}{\partial y} + \hat{\mathbf{k}}\frac{\partial}{\partial z}$$

A. Gradient of a Scalar Field ($\nabla\phi$) & Directional Derivative:

- **Gradient:** $\nabla\phi = \hat{\mathbf{i}}\frac{\partial\phi}{\partial x} + \hat{\mathbf{j}}\frac{\partial\phi}{\partial y} + \hat{\mathbf{k}}\frac{\partial\phi}{\partial z}$ (normal to level surface $\phi(x, y, z) = c$).
- **Unit Normal Vector:** $\hat{\mathbf{n}} = \frac{\nabla\phi}{|\nabla\phi|}$.
- **Directional Derivative along $\vec{a}$:** $\text{D.D.} = \nabla\phi \cdot \frac{\vec{\mathbf{a}}}{|\vec{\mathbf{a}}|}$ (maximum magnitude = $|\nabla\phi|$ in direction of $\nabla\phi$).

B. Divergence ($\nabla \cdot \vec{F}$) & Curl ($\nabla \times \vec{F}$):

- **Divergence (Scalar):** $\text{div}\,\vec{\mathbf{F}} = \nabla \cdot \vec{\mathbf{F}} = \frac{\partial \mathbf{F_1}}{\partial x} + \frac{\partial \mathbf{F_2}}{\partial y} + \frac{\partial \mathbf{F_3}}{\partial z}$.

- **Solenoidal Field:** $\nabla \cdot \vec{\mathbf{F}} = \mathbf{0}$.

- **Curl (Vector):** $\text{curl}\,\vec{\mathbf{F}} = \nabla \times \vec{\mathbf{F}} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$.

- **Irrotational (Conservative) Field:** $\nabla \times \vec{\mathbf{F}} = \vec{\mathbf{0}} \iff \vec{\mathbf{F}} = \nabla\phi$.

University Question (6 Marks - MMDU Question Paper):
Find the directional derivative of $\phi = x^2yz + 4xz^2$ at point $P(1,-2,-1)$ in the direction of vector $\vec{a} = 2\hat{i} - \hat{j} - 2\hat{k}$.

Step 1 🚀: Calculate $\nabla\phi$ at $P(1,-2,-1)$

$$\nabla\phi = (2xyz + 4z^2)\hat{i} + (x^2z)\hat{j} + (x^2y + 8xz)\hat{k}\Big|_{(1,-2,-1)} = 8\hat{i} - \hat{j} - 10\hat{k}$$

Step 2 ⚡: Project onto Unit Vector $\hat{a}$

$$\hat{a} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{\sqrt{4+1+4}} = \frac{2\hat{i} - \hat{j} - 2\hat{k}}{3} \implies \text{D.D.} = \nabla\phi \cdot \hat{a} = \frac{16 + 1 + 20}{3} = \frac{37}{3}$$

🎯 Directional Derivative: $\frac{37}{3} \approx \mathbf{12.33}$.

University Question (6 Marks - MMDU Question Paper):
Prove that $\vec{F} = (y^2\cos x + z^3)\hat{i} + (2y\sin x - 4)\hat{j} + (3xz^2 + 2)\hat{k}$ is irrotational, and find its scalar potential ϕ.

Step 1 🚀: Evaluate curl $\vec{F} = \nabla \times \vec{F}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2\cos x + z^3 & 2y\sin x - 4 & 3xz^2 + 2 \end{vmatrix} = (0)\hat{i} - (3z^2 - 3z^2)\hat{j} + (2y\cos x - 2y\cos x)\hat{k} = \vec{0}$$

Step 2 ⚡: Integrate Exact Differential $d\phi = \vec{F} \cdot d\vec{r}$

$$d\phi = d(y^2\sin x) + d(xz^3) - d(4y) + d(2z) \implies \phi = y^2\sin x + xz^3 - 4y + 2z + C$$

🎯 $\nabla \times \vec{F} = \vec{0}$ (Irrotational) | Scalar Potential: $\phi = \mathbf{y^2\sin x + xz^3 - 4y + 2z + C}$.

University Solved Examination Questions (Unit 3)

MMDU Official University Solved Examination Questions (Unit 3)

Course: Mathematics I (`BMAT-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 7 (1 Mark): Homogeneous Degree via Euler's Theorem

Question:

Find the degree of homogeneity for $u(x,y) = \frac{x^3+y^3}{x-y}$ and evaluate $xu_x + yu_y$.

Step-by-Step Solution:

Since $u(tx,ty) = \frac{t^3(x^3+y^3)}{t(x-y)} = t^2u(x,y)$, the degree is $n = 2$. By Euler's theorem, $xu_x + yu_y = 2u$.

🎯 **Verified Result:** Degree $n = 2$ | $xu_x + yu_y = 2u$.

📘 Question 19 (4 Marks): Two-Variable Extrema Determination

Question:

Find and classify all local extrema and saddle points of $f(x,y) = x^3 + y^3 - 3axy$ $(a > 0)$.

Step-by-Step Solution:

Stationary system: $f_x = 3x^2 - 3ay = 0$ and $f_y = 3y^2 - 3ax = 0 \implies$ critical points $(0,0)$ and (a,a).

With $r = 6x, s = -3a, t = 6y$ and $\Delta = rt - s^2 = 36xy - 9a^2$:

- At $(0,0)$: $\Delta = -9a^2 < 0 \implies$ **Saddle Point**.
- At (a,a): $\Delta = 27a^2 > 0, r = 6a > 0 \implies$ **Local Minimum** with $f_{\min} = -a^3$.

🎯 **Verified Result:** $(0,0)$ is a Saddle Point; (a,a) is a Local Minimum with value $-a^3$.

📘 Question 23 (6 Marks): Constrained Optimization via Lagrange Multipliers

Question:

Find the maximum and minimum distances from the origin to the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ $(a > b > c > 0)$.

Step-by-Step Solution:

Using $F = (x^2 + y^2 + z^2) + \lambda\left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1\right)$, setting $F_x = F_y = F_z = 0$ yields $\lambda = -a^2, -b^2, -c^2$, corresponding to distances $d_{\max} = a$ and $d_{\min} = c$.

🎯 **Verified Result:** Maximum distance $= a$ | Minimum distance $= c$.

UNIT 4 • MMDU MULLANA ACADEMIC CURRICULUM

Unit 4: Unit IV: Sequences & Series (05 Hrs) and Fourier Series (05 Hrs)

Exam Weightage: 15 Marks (3×1M + 1×2M + 1×4M + 1×6M) • **Estimated Study Time:** 40 mins

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1. Convergence Tests: Comparison, Ratio, Raabe's & Logarithmic

1. In-Depth Theory of Infinite Series Convergence & Divergence

Let $\sum_{n=1}^{\infty} u_n = u_1 + u_2 + \cdots + u_n + \ldots$ be an infinite series with partial sums $S_k = \sum_{n=1}^{k} u_n$.

- **Necessary Condition (n^{th} Term Test):** If $\sum u_n$ converges, then $\lim_{\mathbf{n}\to\infty} \mathbf{u_n} = \mathbf{0}$. If $\lim_{n\to\infty} u_n \not\equiv 0$, the series **strictly diverges**.

- **Geometric Series $\sum r^n$:** Converges iff $|r| < 1$.

- **Harmonic p-Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$:** Converges strictly if $\mathbf{p > 1}$, and diverges if $\mathbf{p \leq 1}$.

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2. Systematic Positive-Term Convergence Tests Pipeline

1. **Limit Comparison Test:** If $\lim_{n\to\infty} \frac{u_n}{v_n} = l$ $(0 < l < \infty$, finite and non-zero), then $\sum u_n$ and $\sum v_n$ converge or diverge together.

2. **D'Alembert's Ratio Test:** Let $\mathbf{L} = \lim_{\mathbf{n}\to\infty} \frac{\mathbf{u_n}}{\mathbf{u_{n+1}}}$:

- $\mathbf{L > 1} \implies$ **Convergent** | $\mathbf{L < 1} \implies$ **Divergent** | $\mathbf{L = 1} \implies$ **Test Fails** (use Raabe's or Logarithmic test).

3. **Raabe's Test (when $L = 1$):** Evaluate $\mathbf{R} = \lim_{\mathbf{n}\to\infty} \mathbf{n}\left(\frac{\mathbf{u_n}}{\mathbf{u_{n+1}}} - \mathbf{1}\right)$:

- $\mathbf{R > 1} \implies$ **Convergent** | $\mathbf{R < 1} \implies$ **Divergent** | $\mathbf{R = 1} \implies$ **Test Fails**.

4. **Logarithmic Test (when $L = 1$ with exponents/e):** Evaluate $\mathbf{L'} = \lim_{\mathbf{n}\to\infty} \mathbf{n} \ln\left(\frac{\mathbf{u_n}}{\mathbf{u_{n+1}}}\right)$:

- $\mathbf{L' > 1} \implies$ **Convergent** | $\mathbf{L' < 1} \implies$ **Divergent**.

University Question (6 Marks - Classic MMDU Exam Problem):
Examine the convergence of the series:

$$\frac{1}{2} + \frac{1 \cdot 3}{2 \cdot 4} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} + \cdots + \frac{1 \cdot 3 \cdot 5 \ldots (2n-1)}{2 \cdot 4 \cdot 6 \ldots (2n)} + \ldots$$

Step 1 🚀: Ratio Test Setup
Here $\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1} \implies L = \lim_{n\to\infty} \frac{2n+2}{2n+1} = 1 \implies$ **Ratio Test Fails!**

Step 2 ⚡: Apply Raabe's Test

$$R = \lim_{n\to\infty} n\left(\frac{2n+2}{2n+1} - 1\right) = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1$$

🎯 Ratio test fails ($L = 1$); Raabe's Test yields $R = \frac{1}{2} < 1 \implies$ Series is **Divergent**.

2. Cauchy's Root Test, Alternating Series & Leibnitz's Rule

1. Cauchy's Root Test, Alternating Series & Leibnitz's Rule

A. Cauchy's n^{th} Root Test:

✅ **FUNDAMENTAL LAW / RULE:**
Theorem: Let $\sum u_n$ be a positive-term series and evaluate $\mathbf{L = \lim_{n\to\infty}(u_n)^{1/n}}$:

- $\mathbf{L < 1} \implies$ **Convergent** | $\mathbf{L > 1} \implies$ **Divergent** | $\mathbf{L = 1} \implies$ **Test Fails**.

B. Alternating Series & Leibnitz's Rule:

✅ **FUNDAMENTAL LAW / RULE:**
Theorem: An alternating series $\sum_{n=1}^{\infty}(-1)^{n-1}u_n = u_1 - u_2 + u_3 - u_4 + \ldots$ $(u_n > 0)$ converges if:

1. **Monotonic Decrease:** $u_{n+1} \leq u_n$ for all n.

2. **Vanishing Limit:** $\lim_{n\to\infty} u_n = 0$.

C. Absolute vs Conditional Convergence:

- **Absolutely Convergent:** $\sum |u_n|$ converges (e.g., $\sum \frac{(-1)^{n-1}}{n^2}$).
- **Conditionally Convergent:** $\sum u_n$ converges via Leibnitz's test, but $\sum |u_n|$ diverges (e.g., $\sum \frac{(-1)^{n-1}}{n}$ and $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$).

University Question (4 Marks - MMDU Question Paper):
Test the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{\sqrt{n}}$ for absolute and conditional convergence.

Step 1 🚀: Verify Leibnitz's Conditions
Here $u_n = \frac{1}{\sqrt{n}} > 0$. Since $\sqrt{n+1} > \sqrt{n} \implies u_{n+1} < u_n$, and $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$, the series **converges** by Leibnitz's Rule.

Step 2 ⚡: Check Absolute Series $\sum |u_n|$

$$\sum_{n=1}^{\infty} |u_n| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$$

Since $p = \frac{1}{2} \leq 1$, $\sum |u_n|$ **diverges**. Therefore, the series is **Conditionally Convergent**.

🎯 $\sum \frac{(-1)^{n-1}}{\sqrt{n}}$ is **Conditionally Convergent** | $\sum \frac{(-1)^{n-1}}{n^2}$ is **Absolutely Convergent**.

3. Fourier Series: Euler's Formulae & Discontinuity

1. Fourier Series: Foundations & Euler's Formulas

Fourier Representation in $(-\pi, \pi)$ or $(0, 2\pi)$:

$$\mathbf{f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]}$$

$$\mathbf{a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx, \quad a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\,dx, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx)\,dx}$$

Dirichlet's Conditions & Jump Discontinuity:

At any finite jump discontinuity $x = c$, the Fourier series converges to the average of the left and right limits:

$$\mathbf{S(c) = \frac{f(c^+) + f(c^-)}{2}}$$

University Question (6 Marks - MMDU Question Paper):
Obtain the Fourier series for $f(x) = x^2$ in $(-\pi, \pi)$. Hence deduce $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Step 1 🚀: Symmetry & Coefficients
Since $f(-x) = (-x)^2 = x^2 = f(x)$, $f(x)$ is **even** $\implies b_n = 0$.

$$a_0 = \frac{2}{\pi} \int_0^{\pi} x^2\,dx = \frac{2\pi^2}{3}, \quad a_n = \frac{2}{\pi} \int_0^{\pi} x^2 \cos(nx)\,dx = \frac{4(-1)^n}{n^2}$$

Step 2 ⚡: Fourier Expansion & Deduction at $x = \pi$

$$x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(nx) \implies \text{At } x = \pi: \ \pi^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{1}{n^2} \implies \sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

🎯 Fourier Series: $x^2 = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}\cos(nx)$ | Deduced: $\sum_{\mathbf{n=1}}^{\infty} \frac{\mathbf{1}}{\mathbf{n^2}} = \frac{\pi^2}{6}$.

4. Even/Odd Symmetries, Change of Interval & Half-Range Series

1. Even/Odd Symmetries, Change of Interval $(-l, l)$ & Half-Range Series

Parity / Domain	Fourier Coefficients	Series Expansion
Even Function in $(-l, l)$	$b_n = 0,\ a_0 = \frac{2}{l}\int_0^l f(x)dx,\ a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$	$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right)$
Odd Function in $(-l, l)$	$a_0 = 0, a_n = 0,\ b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$	$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right)$
Half-Range Cosine in $(0, l)$	$b_n = 0,\ a_0 = \frac{2}{l}\int_0^l f(x)dx,\ a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}dx$	Even periodic extension of period $2l$
Half-Range Sine in $(0, l)$	$a_0 = 0, a_n = 0,\ b_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}dx$	Odd periodic extension of period $2l$

University Question (6 Marks - MMDU Question Paper):
Find the Fourier series for the triangular wave $f(x) = l - |x|$ in the arbitrary interval $(-l, l)$. Hence deduce $\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{8}$.

Step 1 🚀: Symmetry Check in $(-l, l)$
Since $f(-x) = l - |-x| = f(x)$, $f(x)$ is **even** $\implies b_n = 0$ and $a_0 = \frac{2}{l}\int_0^l (l - x)dx = l$.

Step 2 ⚡: Compute a_n **and Deduce at** $x = 0$

$$a_n = \frac{2}{l}\int_0^l (l - x)\cos\left(\frac{n\pi x}{l}\right)dx = \frac{2l}{n^2\pi^2}[1 - (-1)^n] = \begin{cases} \frac{4l}{n^2\pi^2}, & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$$

At $x = 0$: $l = \frac{l}{2} + \frac{4l}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \ldots\right) \implies \sum_{n=1,3,5\ldots}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{8}$.

🎯 Fourier Series: $\mathbf{l - |x|} = \frac{\mathbf{l}}{2} + \frac{\mathbf{4l}}{\pi^2}\sum_{\mathbf{n=1,3,5\ldots}} \frac{\mathbf{1}}{\mathbf{n^2}}\cos\left(\frac{\mathbf{n\pi x}}{\mathbf{l}}\right)$ | Deduced: $\sum_{\mathbf{n=1,3,5\ldots}} \frac{\mathbf{1}}{\mathbf{n^2}} = \frac{\pi^2}{8}$.

University Solved Examination Questions (Unit 4)

MMDU Official University Solved Examination Questions (Unit 4)

Course: Mathematics I (`BMAT-001`)

Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (4 Marks Short Long), D (6 Marks Long)

📘 Question 10 (1 Mark): Failure of Ratio Test on p-Series

Question:
Why does D'Alembert's Ratio Test fail on the p-series $\sum \frac{1}{n^p}$?

Step-by-Step Solution:
Because $L = \lim_{n\to\infty} \frac{u_n}{u_{n+1}} = \lim_{n\to\infty} (1+\frac{1}{n})^p = 1^p = 1$ for every p, so the Ratio Test provides no conclusion and Raabe's Test must be used.

🎯 **Verified Result:** $L = 1$ for all p, hence Ratio Test fails.

📘 Question 20 (4 Marks): Raabe's Test on Higher-Order Series

Question:
Test the convergence of $1 + \frac{1}{2}\cdot\frac{1}{3} + \frac{1\cdot3}{2\cdot4}\cdot\frac{1}{5} + \frac{1\cdot3\cdot5}{2\cdot4\cdot6}\cdot\frac{1}{7} + \dots$

Step-by-Step Solution:
Here $\frac{u_n}{u_{n+1}} = \frac{2n+2}{2n+1}\cdot\frac{2n+3}{2n+1} = \frac{4n^2+10n+6}{4n^2+4n+1} \to 1$ (Ratio Test fails).
Applying Raabe's Test:

$$R = \lim_{n\to\infty} n\left(\frac{u_n}{u_{n+1}} - 1\right) = \lim_{n\to\infty} \frac{6n^2+5n}{4n^2+4n+1} = \frac{6}{4} = \frac{3}{2} > 1$$

Hence the series is **Convergent**.

🎯 **Verified Result:** Convergent by Raabe's Test ($R = 1.5 > 1$).

📘 Question 24 (6 Marks): Full Fourier Expansion of Discontinuous Square Wave

Question:

Obtain the Fourier series expansion for $f(x) = -k$ on $(-\pi, 0)$ and $f(x) = +k$ on $(0, \pi)$. Hence deduce $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

Step-by-Step Solution:

Since $f(x)$ is odd, $a_0 = 0, a_n = 0$, and $b_n = \frac{2}{\pi}\int_0^{\pi} k \sin(nx)dx = \frac{2k}{n\pi}[1 - (-1)^n] = \frac{4k}{n\pi}$ for odd n.

$$f(x) = \frac{4k}{\pi}\left(\sin x + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots\right)$$

Substituting $x = \frac{\pi}{2}$ where $f(\pi/2) = k$ yields $1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots = \frac{\pi}{4}$.

🎯 **Verified Result:** $f(x) = \frac{4k}{\pi}\sum_{n=1,3,5\dots} \frac{\sin nx}{n}$ | Gregory-Leibniz series $\frac{\pi}{4}$ proved.